THE CHEMICAL INDUSTRY

INTRODUCTION

German Exploitation of the Synthetic-Chemical Industry

The history of the German chemical industry covers a period of a little less than a century. The foundation on which the whole vast structure of the industry has been built is the preparation of organic substances similar in their properties to natural products and replacing them in their applications. This technique of building up complex organic molecules from simple and readily obtainable chemicals is generally described as the synthetic-chemical industry. In this field, which has become the dominant factor in the world's chemical industry, German chemical firms have shown particular aptitude in developing discoveries generally made outside the firms concerned, and frequently even outside Germany, with the utmost speed, and adapting them to commercial production in a remarkably short space of time.

The first fields of synthetic chemistry to be exploited by the German industry—and those in which they are still pre-eminent—were the manufacture of synthetic drugs and synthetic dyes. The techniques involved in these two fields are closely related, and it repeatedly happens that research work directed originally in one of these fields has produced results of incalculable value in the other. The very foundation stone of synthetic organic chemistry is an excellent example of this close inter-relation between the chemistry of synthetic dyes and the chemistry of synthetic drugs.

In 1856, W. H. Perkin, stimulated by the discovery that the natural alkaloid quinine formed traces of aniline on hydrolysis with alkali, attempted to prepare synthetic quinine from aniline by reversing the process. His researches led, however, not to the production of quinine, but to a violet dye to which the name "mauvein" was given. The importance of this first synthesis of a complex organic molecule was quickly appreciated in Germany, and the rapid developments made there in the field of synthetic dyes, closely followed by the development of synthetic drugs prepared by similar techniques, soon gave to the German chemical industry an advantage over its less far-sighted competitors in other lands which it holds to the present time, and which was never seriously challenged anywhere until 1914.

Perkin's basic discovery was immediately developed by the Farben-fabrik Friedrich Bayer & Co., of Elberfeld, founded in 1850, and by the Badische Anilin und Soda-Fabrik, of Ludwigshafen-Mannheim, founded in 1861. These pioneers were quickly followed by the Farbwerke Meister Lucius und Brüning, of Hoechst, and the Aktiengesellschaft für Anilin Fabrikation. These pioneer firms were followed by the Farbwerke Dr. E. ter Meer & Co., of Uerdingen, founded in 1877. This firm never achieved the importance of the four founders of the synthetic dyestuffs industry, and gradually became more and more under their influence until its complete absorption in 1916. The immediate and striking success of the German chemical firms concentrating on synthetic organic chemicals stimulated the developments of the German chemical industry in the wider field of general organic and heavy chemicals. Prominent in this field were the firms of Kalle & Co., of Biebrich, and the Chemische Fabrik Griesheim Elektron, of Frankfort, both founded in 1863. At the same time, the old-established firm of Leopold Casella, of Frankfort, founded as long ago as 1815 for the importation and treatment of natural dyestuffs, gradually became interested in the preparation of synthetic colours and made some developments in this field.

The Firm of Bayer and Synthetic Medicaments

From its inception the firm of Friedrich Bayer paid particular attention to the preparation of synthetic medicaments, and gradually developed this side of their business more than their other activities. The importance of Bayer as a pharmaceutical firm, rather than as a producer of dyestuffs, dates from the discovery of *phenacetin* in 1888. This drug was first prepared in the dyestuffs research laboratories at Elberfeld, and the company were quick to realise the importance of this, the first synthetic preparation having analgesic and anti-pyretic properties. Commercial production was started immediately, and the striking success of phenacetin in treating the severe influenza epidemic which swept Europe in 1889, saw the Bayer concern firmly established as a manufacturer of synthetic pharmaceuticals in a little more than eighteen months. Almost immediately after the discovery of phenacetin, a synthetic drug of similar properties, known as *antipyrin*, was introduced, and this again was in time to find successful use in the epidemic of 1889. From this date Bayer never ceased to concentrate on the preparation of medical specialities, and by liberally subsidising medical research in laboratories throughout Germany, they made available to themselves the products of the best discoveries of the period. The Bayer directors quickly realised that synthetic drugs alone could not cover the whole field of medicine, and their active support of the fundamental pathological researches of Koch and Emil von Behring led to the formation of a special serum department where standardised sera were prepared and made available to medical and veterinary workers as "Bayer" products. The most important immediate results of this development were the introduction of anti-diphtheria serum in 1894 and of tuberculin shortly afterwards. Many other anti-toxins, notably against tetanus, anthrax and botulism, were developed in succeeding years.

The expansion of Germany's imperialist ambitions turned the attention of the Bayer Research Laboratories to the field of tropical medicine at the end of the first decade of the twentieth century. The first results of this drive became apparent in the early years of the 1914/18 war, with the appearance of *Germanin* (Bayer 205), *Trypaflavine*, *Rivanol*, etc., and culminated in the discovery of the anti-malarials, *plasmoquin* and *atebrin*, in 1920. From that date to the present war a host of new preparations, particularly new mixtures based on known synthetic drugs, have been launched by Bayer, culminating in the discovery of *Dolantin*, the first synthetic substitute for morphine, in 1939.

The development of the other important pharmaceutical manufacturers in Germany has, in greater or less degree, followed the same course as the development of the Bayer organisation.